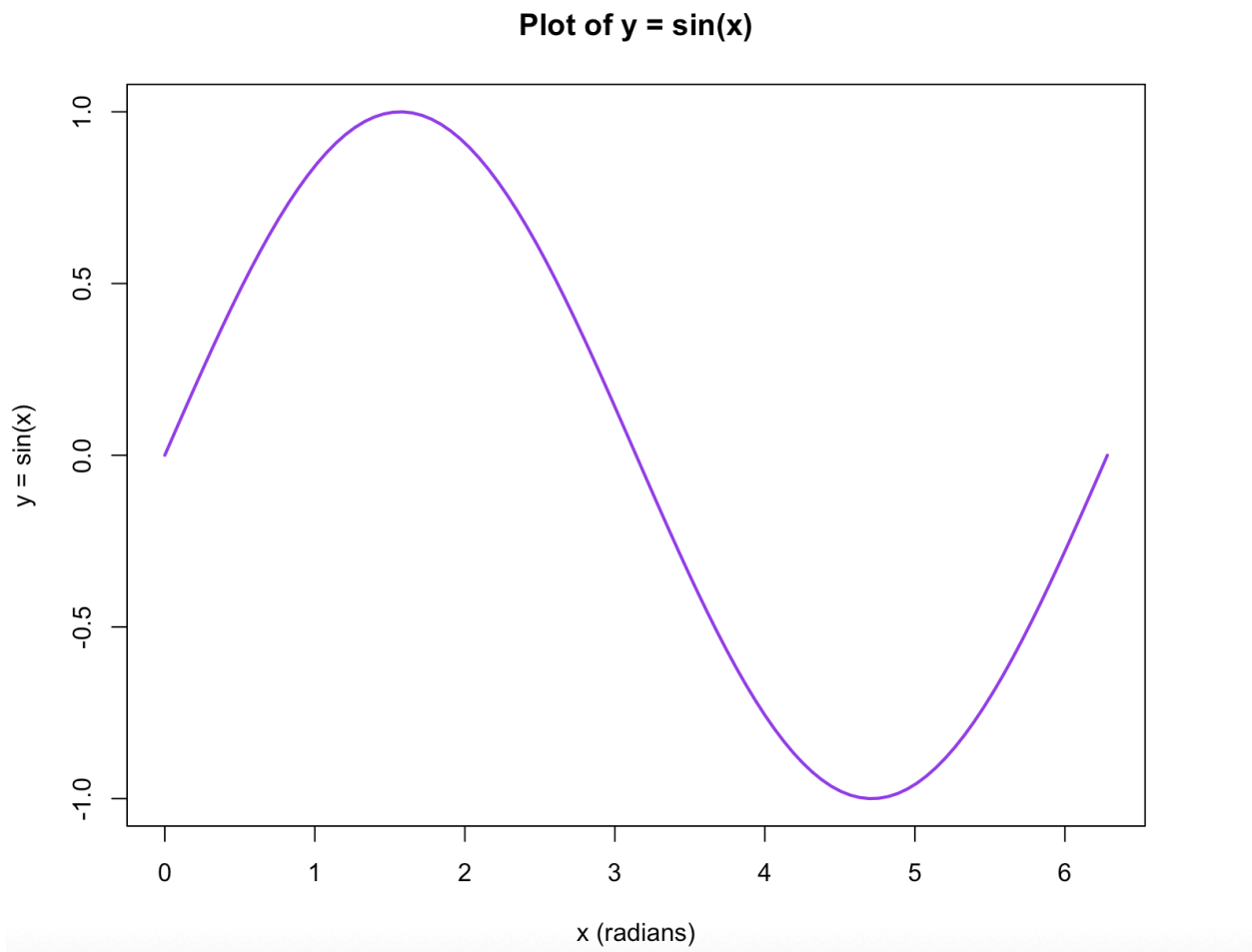
Question 1)

Plot y = sin(x) with axis labels and chart title



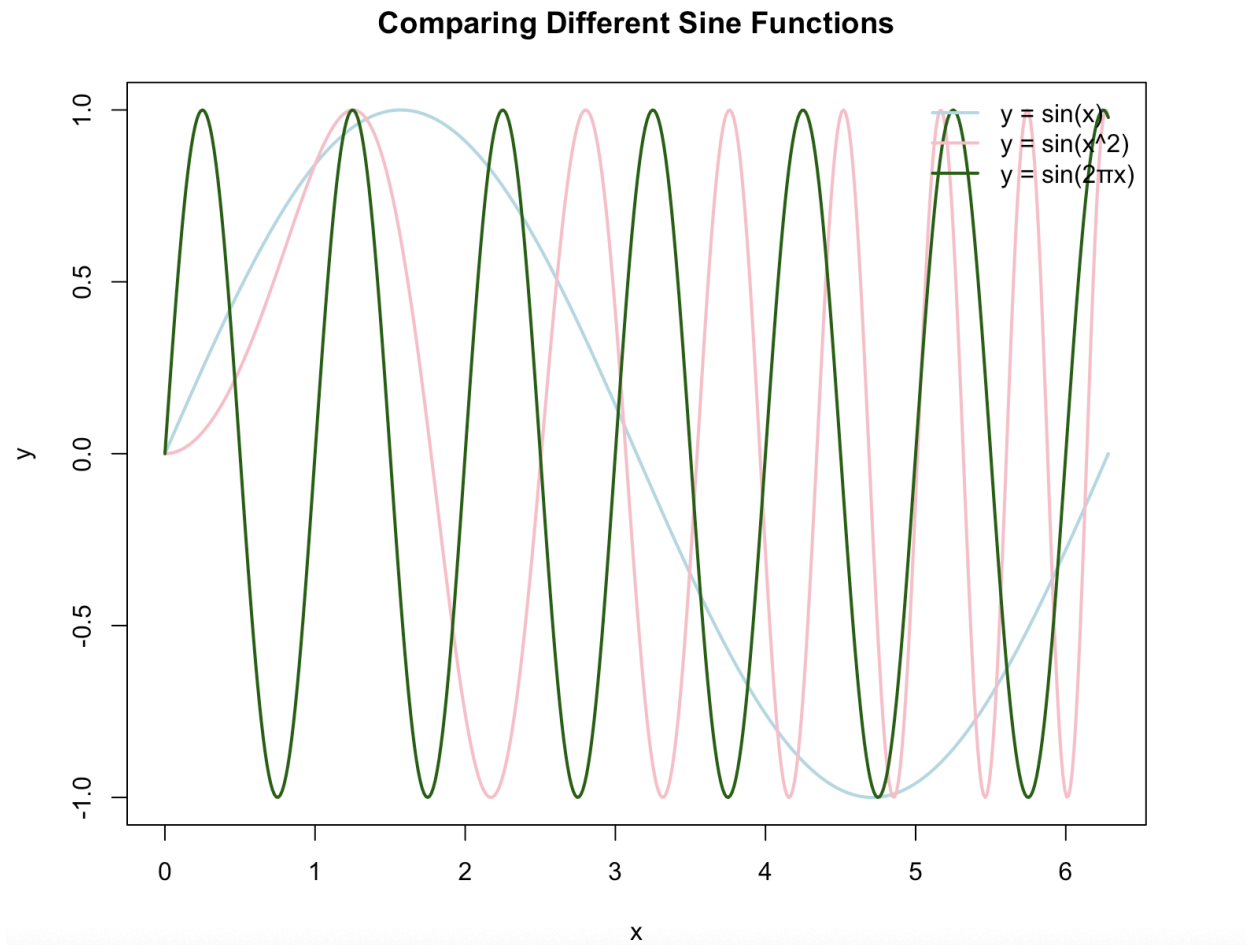
Graph: 01

This graph, y = sin(x), is the simplest mathematical example of a periodic oscillation. Note, if we want to convert the x-axis to degrees, we apply the following formula: Degrees = radians * (180/pi).

- An Example for the function y = sin(x) is the motion of a pendulum

Question 2)

*Hold the previous plot and insert the functions y = sin(x^2) & y = sin(2*pi*x). Label the functions inside the figure.*



Graph: 02

Upon laying the functions y = sin(x^2), and y = sin(2*pi*x) on top of the original function y = sin(x) we can see that all three functions have different frequencies. We can see that the function y = sin(2*pi*x) has the shortest wavelength, completing a full oscillation in 1 time period (T = 2pi/2pi = 1), whereas the function y = sin(x^2) does not have a fixed single period because as x increases the duration of a full oscillation decreases.

- An example for the function y = sin(x^2) are Chirp signals in radar and sonar. When frequencies increase over time to scan environments.
- An example for the function y = sin(2*pi*x) is when a wheel turns one revolution per unit distance

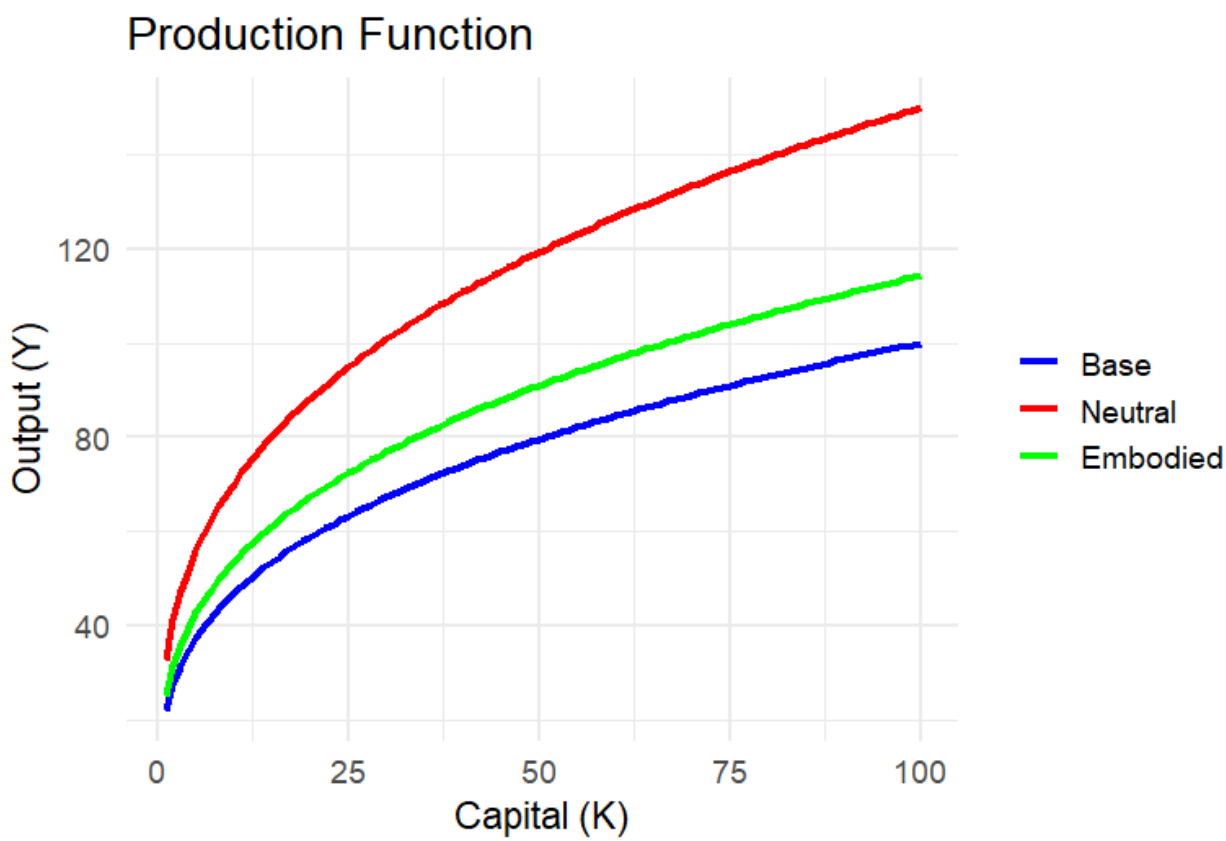
Question 3)

Plot the production function $Y = K^{\alpha}L^{1-\alpha}$ in 2d.

1. What happens when technology (A) is inserted in a neutral way? Plot and analyse the function

2. What happens when technology is embodied in capital? plot and analyse the function

Keeping labour fixed :



Graph: 03

The above graph shows the following :

- Blue (Base): Standard Cobb–Douglas without technology.
- Dashed Red (Neutral A): Shifts output uniformly upward.
- Dotted Green (Embodied A in K): Increases faster at high levels of capital.

Interpretation:

- At low capital, neutral A gives higher output than embodied A.

- At high capital, embodied A catches up and may surpass neutral A, because A is compounding with K inside the exponent.

So, neutral technology is better overall, especially when:

- Labour is fixed and not too scarce.
- Capital isn't very high yet.

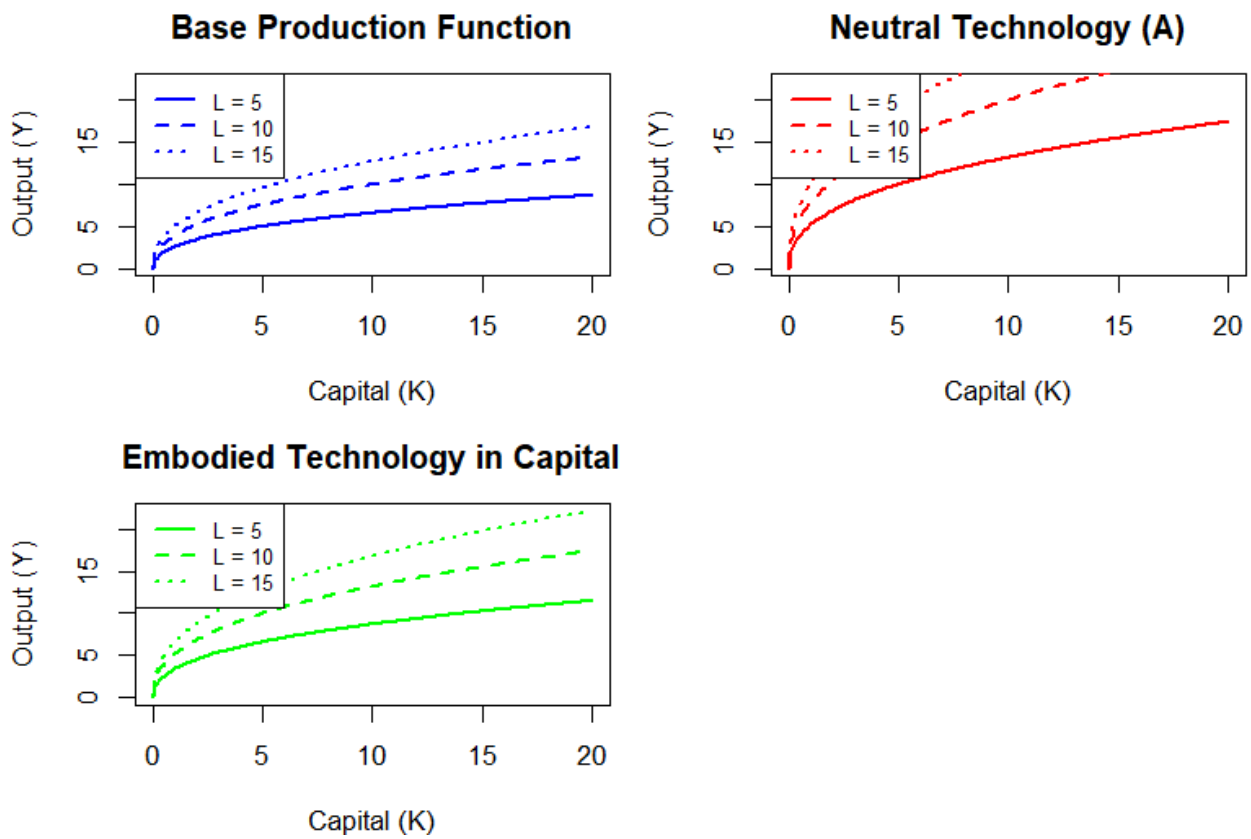
But if an economy is heavily capital-intensive, embodied tech can eventually outperform.

The above graph represents a Cobb Douglas production function where the output is determined by capital and labour.The base curve represents ($Y=K^{\alpha} L^{1-\alpha}$)when labour is fixed.This indicates that the total output will rise as the capital rises, but marginal output decreases with each extra unit of capital.

The production function becomes $Y=A(K^{\alpha} L^{1-\alpha})$ with the introduction of neutral (Hicks-neutral) technological progress. Consequently, the whole curve rises and output rises proportionately in all capital amounts. The curve's shape remains constant, suggesting that diminishing returns are still relevant. Thus, neutral technology is a general increase in efficiency for all inputs.

On the other hand, the function becomes $Y=(AK)^{\alpha} L^{1-\alpha}$ when technology is embodied in capital. In this case, technological advancement only directly benefits capital. The resulting curve is below the neutral technology case but above the base case. This illustrates how, because labour is unaffected, embodied progress increases output less than neutral progress.

Overall, the comparison reveals two important conclusions: first, advancements in technology cause the production function to shift outward, increasing potential output; and second, the manner in which technology enters the function is important.Neutral progress will lead to more gains in total output while embodied progress results in capital focused gain.



Graph: 04

The above graph shows three cases of the Cobb–Douglas production function:

1. Base case: $Y=K^{\alpha}L^{1-\alpha}$ (Output depends only on K and L.)
2. Neutral technology: $Y=A\,K^{\alpha}L^{1-\alpha}$ (Technology (A) multiplies the entire function.)

- Effect: For any given K,L, output is proportionally higher.
- Both capital and labour benefit equally.

3. Capital-embodied technology:
   $Y=(AK)^{\alpha}L^{1-\alpha}$ (Technology works only through capital.)

- Effect: If capital is scarce, the benefit from A is limited.
- The productivity boost is biased towards capital accumulation.

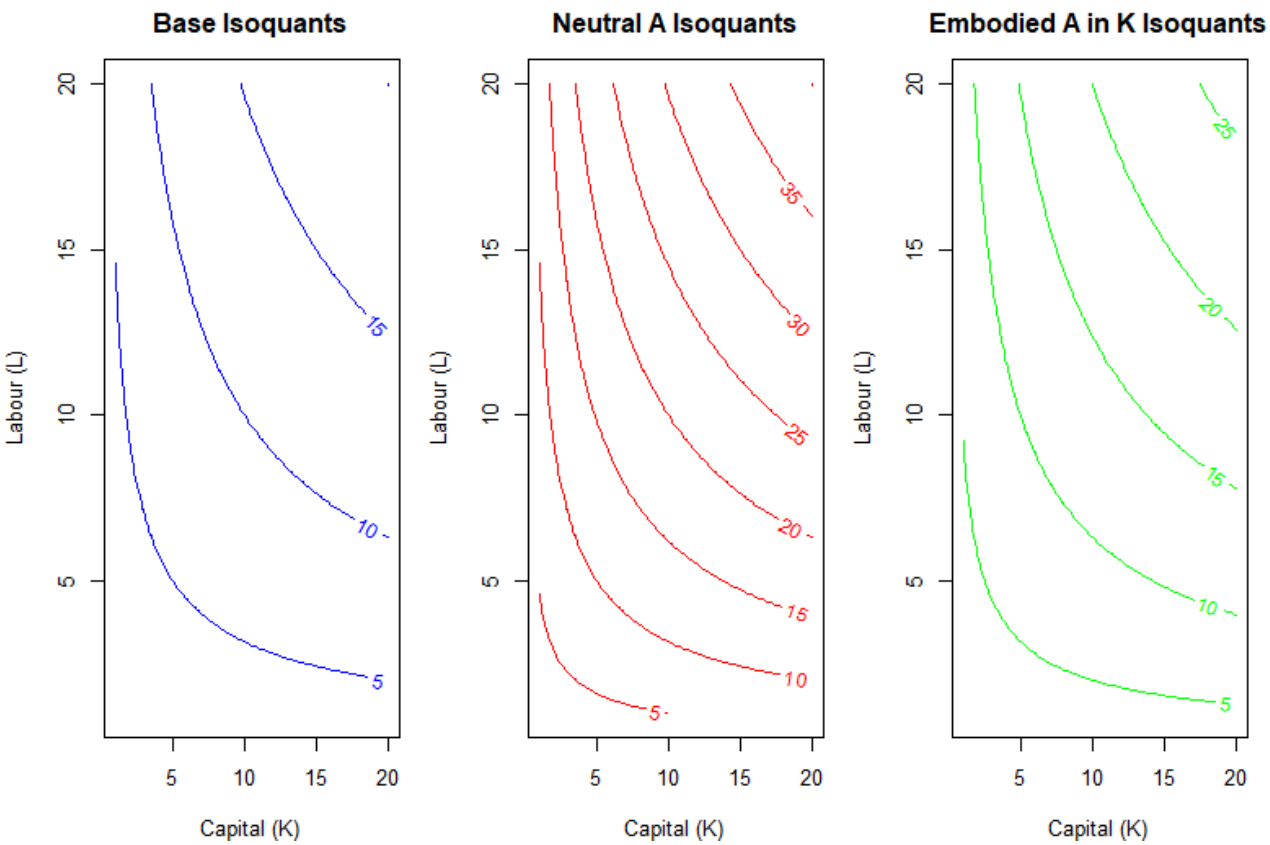
If both labour and capital are present, neutral technology gives a larger and more uniform boost, every input is more productive.

If the economy is very capital-intensive, embodied technology can actually give bigger gains, because A directly amplifies K.

But if the economy is labour-abundant but capital-scarce (like many developing countries), embodied technology is less effective, since without much K, the effect of A doesn't materialize strongly.

So in general, neutral technology tends to give the "best" output growth, because it lifts productivity of the whole system.
But in capital-rich contexts, embodied technology might look better.

When trying to keep both Labour and Capital varying:



Graph: 05

The above graph shows combinations of K and L that give the same output level Y.
Output Y is fixed, and the curve lives in the K–L plane.

Numeric Example

Set $\alpha = 0.4,\ A = 2,\ Y_0 = 10$. Compute required L for two values of K.

Case 1: $K = 5$

- Base:

$$L_{\text{base}} = \left(\frac{10}{5^{0.4}}\right)^{1/0.6} \approx 15.874.$$

- Neutral:

$$L_{\text{neutral}} = \left(\frac{10}{2 \cdot 5^{0.4}}\right)^{1/0.6} = 5.000.$$

- Embodied:

$$L_{\text{embodied}} = \left(\frac{10}{(2 \cdot 5)^{0.4}}\right)^{1/0.6} = 10.000.$$

Case 2: $K = 10$

- Base:

$$L_{\text{base}} = \left(\frac{10}{10^{0.4}}\right)^{1/0.6} = 10.000.$$

- Neutral:

$$L_{\text{neutral}} \approx 3.1498.$$

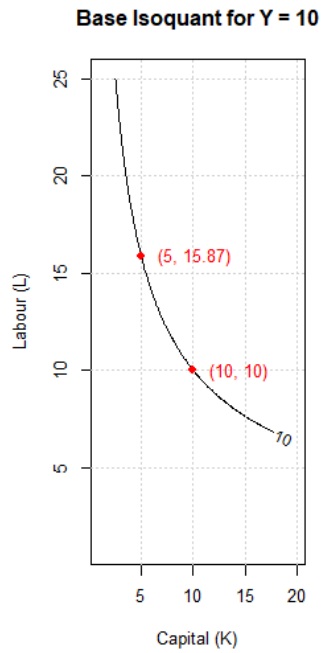
- Embodied:

$$L_{\text{embodied}} \approx 6.2996.$$

Interpretation of numbers: for a given output Y_0, neutral technology reduces required labour the most (here to ≈31.5% of the base when $A = 2$), while embodied technology reduces required labour less (≈63% of base). As capital increases (from 5 to 10), the effect of embodied technology becomes stronger in absolute terms (because embodied tech scales with K), but neutral still reduces L proportionally more for the same A.

Intuition:
Neutral technology A multiplies the whole production function: both capital and labour become more productive. For a given target output it reduces required inputs more than embodied tech of the same magnitude.

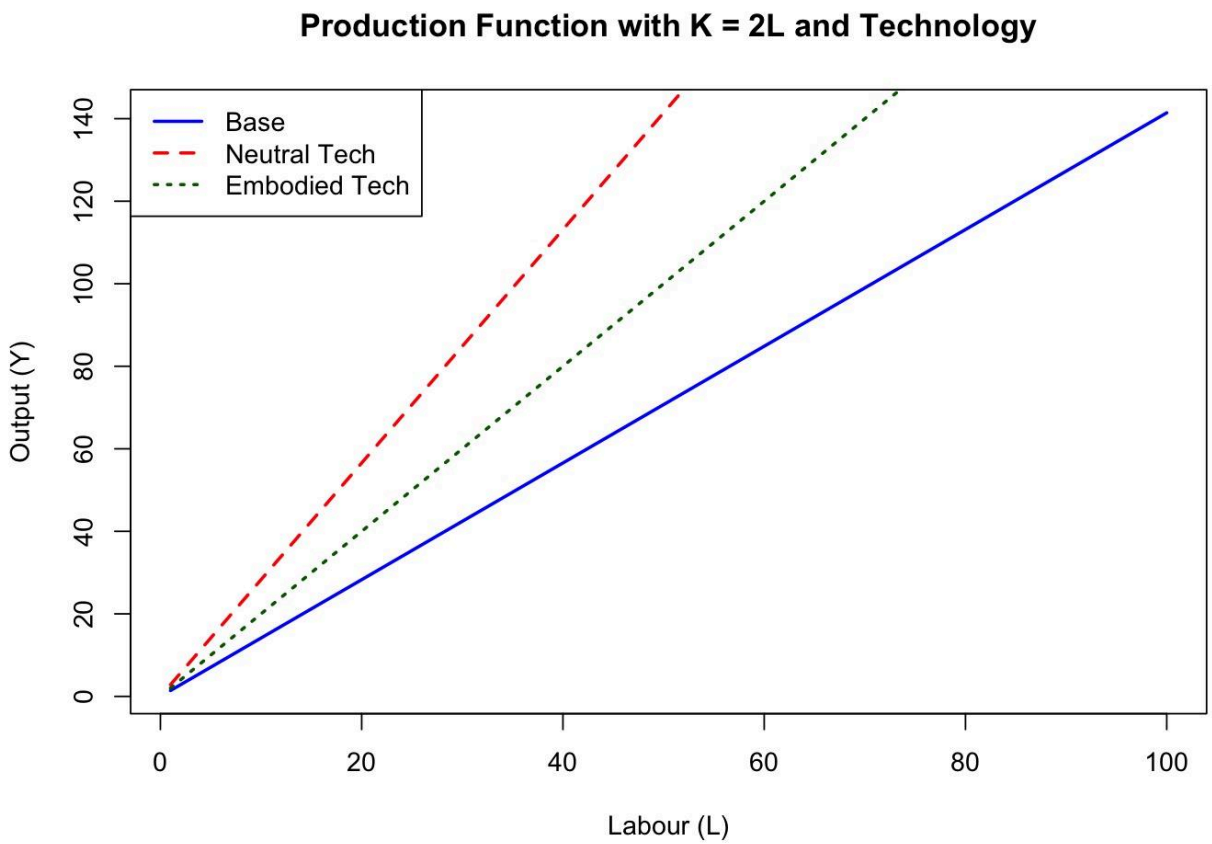
Embodied technology enters only through capital and improves effective capital: gains are biased toward capital and thus are more valuable when the economy is capital-intensive. If capital is scarce (labour-abundant economy), embodied tech yields smaller gains.
Therefore, neutral technology tends to give a larger and more uniform boost across input mixes; embodied technology can outperform neutral tech only in very capital-intensive contexts (large alpha or very high K).



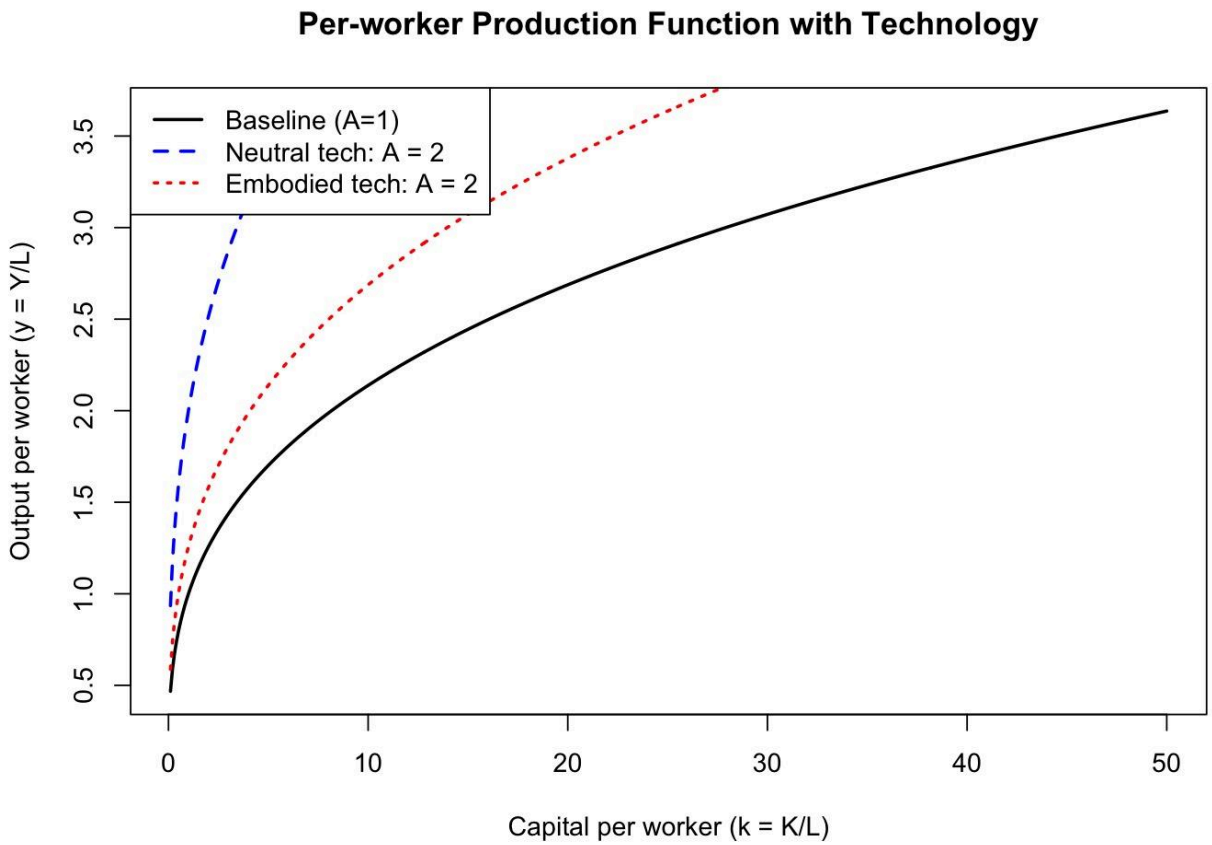
Graph : 06

Hence on the same curve with the same output, changing labour and capital. The base isoquant for Y_0=10 and marked the two points (5, 15.87) and (10, 10).
You can see both points lie on the same curve, so changing K and L appropriately keeps us on the same output level.

WHEN CAPITAL=2L



Graph : 07

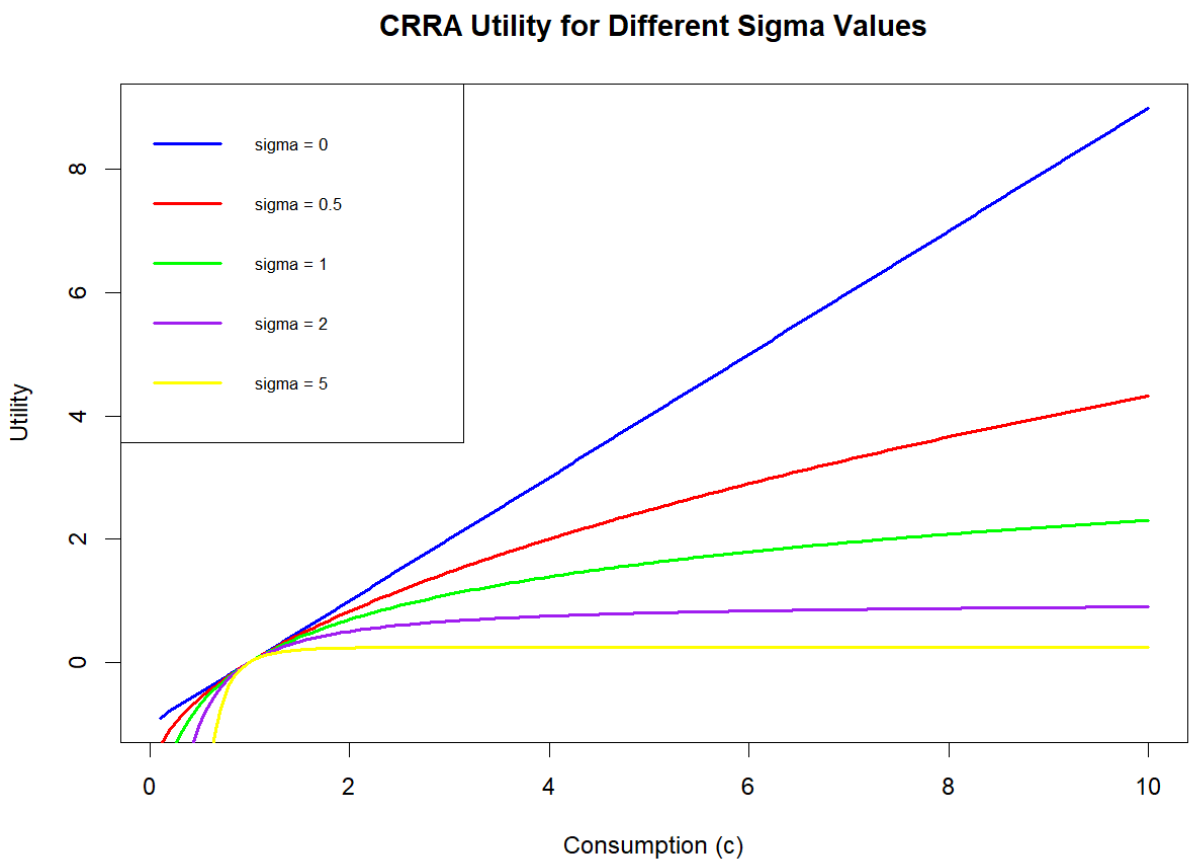


Graph :08

When neutral technology is added it shifts the curve straight up by A,embodied technology shifts the curve A^alpha which will be smaller.Accordingly, the neutral case doubles output per worker if A=2 and α=0.33, but the embodied case only increases it by roughly 25.7%.

Question 4) Plot the standard utility function u = c 1−σ/1 − σ. Experiment the values of Sigma to check whether the utility function satisfies the usual economic assumptions. (Explain what the usual assumptions in the first place)

PLOT OF STANDARD UTILITY FUNCTION : u = c^(1−σ)/1 − σ



Graph : 09

Usual Assumptions about the standard utility function u = c 1−σ/1 − σ are:

1. Monotonicity: More consumption will give more utility ($U'(c) > 0$)
2. Concavity: As consumption increases, marginal utility decreases ($U''(c) < 0$), Utility is strictly concave when $\sigma > 0$, ensuring risk aversion.
3. Risk Aversion: Concavity implies that people prefer smoother consumption paths compared to risky ones.

Observation:

- $\sigma = 0.5$: utility is still increasing, curvature is weak (people are more risk-tolerant).
- $\sigma = 1$: utility becomes log(c).
- $\sigma = 2{,}5$: utility curves bend more strongly → very concave (strong risk aversion, strong diminishing marginal utility).

This confirms the usual assumptions:

- ($U'(c) > 0$), always increasing
- ($U''(c) < 0$), (concave, diminishing marginal utility).

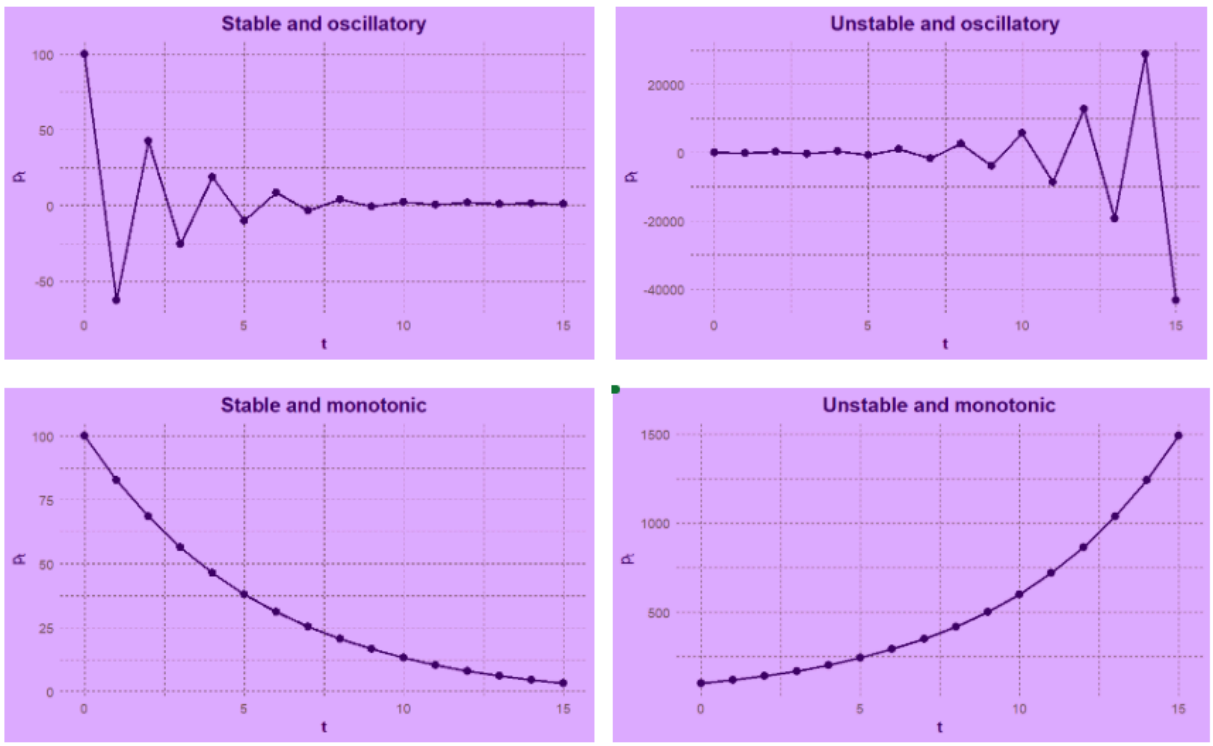
Example:

- $\sigma = 1$ (log utility): People diversify investments (stocks + bonds) but don't avoid risk completely.
- High σ (e.g., 4–5): Someone who always buys insurance, avoids gambling, prefers safe bank deposits.
- Low σ (e.g., 0.5): Someone who speculates in crypto or gambling, because they tolerate risk.

Question 5)

Consider the demand-supply model in the example box on page 12 of Raghavendra and Piiroinen book (An Introduction to Economic Dynamics). Follow the analysis and 1 generate the corresponding figure for different stability conditions described in Table 2.1 in page 26.- four figures for four cases!

STABILITY AND BEHAVIOURAL CONDITIONS



Graph: 10

In order to understand the nature of convergence to the stable equilibrium or the divergence from the fixed point we study the characteristics of the demand and supply slopes.

We know that the demand and supply equations are given as:

Demand: $Qd = d0 + d1*P_t$

Supply: $Qs = s0 + s1*P_{t-1}$

*Note: The supply equation takes into account the previous period's time period (naive expectations)

Parameters:

d0 - constant demand

s0 - constant supply
d1 - slope of the demand curve
s1 - slope of the supply curve
P_t - price at time period t

Stability conditions: (Stable/Unstable)

Stable condition (|s1/d1| < 1)
The ratio |s1/d1| compares how responsive supply is to past prices versus how responsive demand is to current prices.
When |s1/d1| < 1, the price over time converges to an equilibrium price after a number of time periods. Indicating stability. The economic intuition behind being that, If supply reacts less strongly to past prices than demand does to current prices:
→ Each period, the mismatch between demand and supply shrinks.
→ Price oscillations dampen over time and converge to the equilibrium price.

Example scenario: Think of agricultural markets:
Farmers decide how much to plant based on *last season's price*.
If prices were high, they plant more. If prices were low, they cut back.
- Demand is relatively elastic (consumers react strongly to price).
- Supply is relatively inelastic (farmers don't drastically change output due to land and labor constraints).
- Prices stabilize around equilibrium.

Unstable condition (|s1/d1| > 1)
The ratio (|s1/d1|) compares how responsive supply is to past prices versus how responsive demand is to current prices.
When (|s1/d1| > 1), the price over time diverges away from the equilibrium point instead of converging. This indicates instability.

The economic intuition is that, If supply reacts too strongly to past prices compared to how demand reacts to current prices:
→ Each period, the mismatch between demand and supply grows.
→ Price oscillations become larger and move further away from equilibrium.
→ The market fails to settle into a stable equilibrium point.

Example scenario: Agricultural markets again, but where supply response is very elastic: Farmers can quickly and massively increase (or decrease) output if last season's prices were high (or low). Consumers, however, do not react strongly to changes in price (demand is inelastic).
Result: Overproduction after high prices causes a price crash, then underproduction after low prices causes a spike, leading to ever-widening cycles (divergence).

***Knife edge condition (|s1/d1| = 1)**
Here, supply and demand responsiveness are exactly balanced.
Prices do not converge or diverge, they oscillate endlessly around equilibrium with the same amplitude.
This is a fragile, borderline case: any small change in parameters pushes the system toward stability or instability.

Example scenario: A commodity exchange with highly competitive traders. Each round, suppliers adjust output exactly as much as consumers adjust demand. Prices swing back and forth forever, never damping, never exploding.

Behavioural Conditions (Monotonic/Oscillatory)

If (s1 /d1) < 0 it shows oscillatory behavior, if it (s1 /d1) > 0 it shows monotonic behavior. Essentially checking if the value is positive or negative. Why does the sign matter? When it is positive (s1 /d1) > 0 it would mean that both supply and demand slopes "move in the same direction" in the recursion, negative (s1 /d1) < 0 would mean demand and supply “move in opposite direction”.

Oscillatory condition (s1 /d1) < 0
Since demand slopes downward (d1< 0), if the supply slope s1 >0, then the ratio (s1 /d1) < 0.
This gives oscillations around the equilibrium. Instead of smoothly approaching the equilibrium, the system zig-zags across it.
If $|s1/d1|<1$, the oscillations dampen over time, convergent oscillation (stable, but zig-zag).
If $|s1/d1|>1$, the oscillations explode, divergent oscillation (unstable).

Example scenario :
Think of fisheries markets. Fishermen decide their catch for next season based on last season’s prices. If the price was high, they overshoot and bring a huge catch. The price then crashes. Next season, they cut back too much. Price shoots up again. This creates a zig-zagging pattern of prices around the equilibrium.

Monotonic condition (s1 /d1) > 0
If somehow d1>0 (an *upward-sloping demand curve*, which is rare but possible in special cases like Giffen goods or Veblen goods), and since s1>0, then (s1 /d1) > 0
The system adjusts gradually in the *same direction* without oscillating.
If $|s1/d1|<1$, the adjustment is smoothly convergent, equilibrium is reached without overshooting.
If $|s1/d1|>1$, the adjustment is smoothly divergent, prices drift away steadily without zig-zagging.

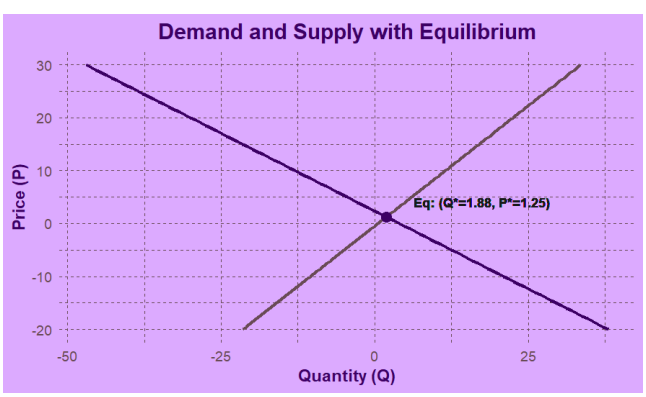
Example scenario :
Consider digital goods markets (like cloud storage or software licenses). If the price last period was high, firms expand server capacity quickly. Consumers, however, react steadily and predictably to current prices. Because supply responds in the same direction as demand pressure, prices move monotonically: they either stabilize smoothly (if response is weak) or diverge monotonically (if response is too strong).

Therefore, Taken together, the stability conditions ($|s1/d1|<1$, $|s1/d1|>1$, or exactly $=1$) and the behavioral conditions ($s1/d1>0$ monotonic vs. $s1/d1<0$ oscillatory) jointly determine how prices evolve in the cobweb model. The magnitude of $|s1/d1|$ tells us whether the system converges to equilibrium (stable), diverges away (unstable), or cycles forever (knife-edge). Meanwhile, the sign of s1/d1 determines the *path* taken: if negative, prices zig-zag around equilibrium (oscillatory adjustment), while if positive, prices move smoothly in one direction (monotonic adjustment). In other words, stability decides *whether* equilibrium is reached, and behavior decides *how* it is reached. Together, they provide a complete picture of the dynamics: markets can stabilize smoothly, stabilize through dampened oscillations, or instead diverge—either monotonically or in ever-widening cycles, depending on the interplay between supply responsiveness and demand sensitivity.

Stable and oscillatory condition demand and supply curve

Taking parameters,
s0 = 0.5
s1 = 1.1
d0 = 4
d1 = -1.7 (same parameters as for the price - time graph above)
$P_{equilibrium}$ = d0-s0/d1-s1 = 4-0.5/1.1-(-1.7) = 1.25
$Q_{equilibrium}=d0+d1P_{eq}=4-1.7\times1.25=1.875$

Here the demand and supply goes through oscillations and dampens to converge to a fixed point making it the stable equilibrium point. Supply reacts less strongly to past prices than demand does to current prices. Also demand slopes downward ($d1< 0$), if the supply slope $s1 >0$, then the ratio $(s1 /d1) < 0$.This gives oscillations around the equilibrium



Graph :11